CS315 · ALGORITHMS ANALYSIS & DESIGN

Mock Midterm Solutions

A practice exam on Lectures 1 to 3 · 90 minutes · 25 marks

Muharib

01 Solution Q1: hash table

Key	5	28	19	15	20	33
k mod 9	5	1	1	6	2	6

(A) Chaining:

Index	Chain
0	∅
1	28 → 19
2	20
3	∅
4	∅
5	5
6	15 → 33
7	∅
8	∅

(B) Double hashing: probe number `j` is `(h1(k) + j·h2(k)) mod 9`.

Key	k mod 9	1 + (k mod 5)	Probes	Cell
5	5	1	5	**5**
28	1	4	1	**1**
19	1	5	1 → 6	**6**
15	6	1	6 → 7	**7**
20	2	1	2	**2**
33	6	4	6 → 1 → 5 → 0	**0**

0	1	2	3	4	5	6	7	8
33	28	20			5	19	15	

```
33:  h1 = 6 taken,  h2 = 1 + (33 mod 5) = 4
```

```
    (6 + 4) mod 9 = 1 taken   →   (6 + 8) mod 9 = 5 taken   →   (6 + 12) mod 9 = 0
```

02 Solution Q2: linear probing

Key	k mod 13	Probes	Cell
20	7	7	**7**
33	7	7 → 8	**8**
7	7	7 → 8 → 9	**9**
46	7	7 → 8 → 9 → 10	**10**
11	11	11	**11**
59	7	7 → 8 → 9 → 10 → 11 → 12	**12**
24	11	11 → 12 → 0	**0**

0	1	2	3	4	5	6	7	8	9	10	11	12
24							20	33	7	46	11	59

b) Key 59 needed 6 probes: cells 7, 8, 9, 10 and 11 were taken, and it landed in 12. Key 24 wrapped around from the end of the table to cell 0.

03 Solution Q3: counting operations

Line	Worst	Best
`currentMin ← A[0]`	2	2
`Temp ← A[0]`	2	2
`k ← 0`	1	1
`for i ← 1 to n − 1 do`	1 + n	1 + n
`if A[i] < currentMin then`	2(n − 1)	2(n − 1)
`currentMin ← A[i]`	2(n − 1)	0
`{ increment counter i }`	2(n − 1)	2(n − 1)
`return currentMin`	1	1
Total	**7n + 1**	**5n + 3**

```
Worst:   2 + 2 + 1 + (1 + n) + 6(n − 1) + 1  =  7n + 1
```

```
Best:    2 + 2 + 1 + (1 + n) + 4(n − 1) + 1  =  5n + 3
```

The worst case here is an array sorted in decreasing order, so every element is smaller than the current minimum.

04 Solution Q4: True / False + MCQ

#	Answer	Reason
1	**a. True**	The slide's own words.
2	**b. False**	$O(n \log n)$ in all cases. Quick sort is the one with an $O(n^2)$ worst case.
3	**b. False**	That defines open addressing. With chaining the item goes into a linked list outside the table.
4	**b**	Divide, then conquer, then combine.
5	**a**	The expected time is $O(1)$, and the worst case is $O(n)$.

05 Solution Q5: Big-Oh

a) $6n^2 + 4n + 8 \leq 6n^2 + 4n^2 + 8n^2 = 18n^2$ for $n \geq 1$

$c = 18, \quad n_0 = 1$

b) $\log n < 3n < n \log n < n^2 < 2^n$

Any other pair that satisfies the inequality is correct, such as $c = 7, n_0 = 6$.

06 Solution Q6: merge sort

Divide	[9 3 7 1 8 2 6 4]			
	[9 3 7 1]		[8 2 6 4]	
	[9 3]	[7 1]	[8 2]	[6 4]
	[9] [3] [7] [1]		[8] [2] [6] [4]	
Merge	[3 9]	[1 7]	[2 8]	[4 6]
	[1 3 7 9]		[2 4 6 8]	
	[1 2 3 4 6 7 8 9]			

b) $T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + T(\text{MERGE})$

c) Levels $= \log(32) + 1 = 5 + 1 = 6$

Level 2 $= 2^2 = 4$ problems, each of size $\frac{32}{4} = 8$

07 Solution Q7: short answers

#	Answer
a	1) Chaining: each cell points to a linked list of the elements mapped there. It is simple but needs memory outside the table. 2) Linear probing: a kind of open addressing that places the item in the next available cell, circularly. 3) Double hashing: a kind of open addressing that jumps by a step given by a second function `d(k)`.
b	When it disperses the keys over the cells in an apparently random, even way, so collisions are few, and it is fast to compute.